SIMPLE CUTS, SIMPLE LANDSCAPE: DIMENSION-FREE RANK BOUNDS FOR BURER–MONTEIRO FACTORIZATIONS VIA RANK-WIDTH

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Abstract. In this work, we study Burer–Monteiro factorizations of semidefinite programs with unit diagonal constraints. For a real symmetric cost matrix whose rank-width over $\mathbb{R}$ is at most k, we prove that every non-flat second-order stationary point is globally optimal whenever the factorization rank is at least $2k$, independently of the matrix size.

1. Introduction

Low-rank matrix factorizations can reduce the number of variables in semidefinite optimization, but introduce nonconvexity. A central question is how large the factorization rank must be for second-order stationarity to guarantee global optimality. In this work, we study semidefinite programs with unit diagonal constraints and ask whether the required rank can be controlled by the structure of the cost matrix, independently of its size.

Let $A = A^\top \in \mathbb{R}^{n\times n}$ have zero diagonal. Consider the semidefinite program

$$\min_{Z=Z^\top \succeq 0} \left\{ -\frac{1}{2}\operatorname{tr}(AZ) : Z_{ii} = 1,\ i = 1,\dots,n \right\}. \tag{1}$$

The zero-diagonal assumption entails no loss of generality, since the diagonal of A contributes only a constant to the objective. Following the Burer–Monteiro approach [BM03], we write $Z = VV^\top$ with $V \in \mathbb{R}^{n\times m}$, thereby restricting the rank of Z to at most m. Positive semidefiniteness is then automatic, and the diagonal constraints require the rows of V to have unit norm. Writing the ith row as $v_i^\top$, we obtain

$$\min_{v_1,\dots,v_n \in \mathbb{S}^{m-1}} E(V), \qquad E(V) = -\sum_{i<j} A_{ij}\langle v_i, v_j\rangle = -\frac{1}{2}\operatorname{tr}(V^\top AV). \tag{2}$$

The parameter m is the factorization rank. Choosing a small m reduces the number of variables, whereas increasing m enlarges the space in which the vectors can move. The optimization landscape therefore depends on both A and m.

General landscape guarantees relate the required factorization rank to the number of constraints. For the problem considered here, a general result implies that, for almost every cost matrix A, every second-order stationary point of (2) yields a global optimum of (1) whenever

$$\frac{m(m+1)}{2} > n$$

[BVB20]. This sufficient condition depends on the number of diagonal constraints, without distinguishing cost matrices according to their interaction structure. Moreover, an almost-everywhere guarantee in the ambient space does not by itself establish a guarantee for every matrix in a prescribed structured family. These observations motivate the following question.

Question. Can we use the structure of the cost matrix to obtain smaller factorization ranks that still guarantee global optimality?

In this work, we study this question through rank-width over $\mathbb{R}$. For a subset $S \subseteq [n]$, the interaction between S and its complement is described by the cut matrix $A[S, [n] \setminus S]$. A rank-decomposition is a tree of maximum degree three whose leaves are identified with $[n]$. Deleting an edge partitions the leaves into two sets and hence determines a cut matrix. The width of the decomposition is the largest rank, over $\mathbb{R}$, of these cut matrices. The real rank-width of A, denoted by $\mathrm{rw}_{\mathbb{R}}(A)$, is the minimum width over all such decompositions. Thus, bounded real rank-width expresses a form of structural simplicity: the interactions across the cuts of a suitable decomposition have low rank, even when the matrix is large or dense.

Main result (informal). If $\mathrm{rw}_{\mathbb{R}}(A) \leq k$ and $m \geq 2k$, then every non-flat second-order stationary point of (2) is a global minimizer of (2).

Here, second-order stationarity means that the Riemannian gradient vanishes and the Riemannian Hessian is positive semidefinite on the tangent space of the product of spheres. The non-flatness condition is specified below, where we state the result in full. For fixed k, the sufficient factorization rank is independent of n. The result applies to each real symmetric cost matrix satisfying the rank-width bound, allows both positive and negative coefficients, and retains the stated non-flatness qualification.

The formulation also connects directly to synchronization. A positive coefficient A_{ij} favors alignment of v_i and v_j, whereas a negative coefficient favors opposite directions; these pairwise preferences need not be simultaneously satisfiable. For $m = 2$, writing $v_i = (\cos\theta_i, \sin\theta_i)$ gives

$$E(V) = -\sum_{i<j} A_{ij} \cos(\theta_i - \theta_j),$$

the phase interaction energy arising in synchronization problems. The factorization rank thus also describes the dimension of the interacting unit vectors. From this perspective, our result relates the structure of the interactions to the dimension sufficient to rule out nonglobal, non-flat second-order stationary points.

1.1. **Main result.** Our main result shows that a factorization rank of at least $2k$ suffices when the cost matrix has rank-width at most k.

Theorem 1.1. *Let $k \geq 1$, and let $A \in \mathbb{R}^{n\times n}$ be symmetric with zero diagonal and rank-width over $\mathbb{R}$ at most k. If $m \geq 2k$, every non-flat second-order stationary point of (2) is a global minimizer.*

1.2. **Related work.** Although semidefinite programs are convex, directly optimizing over a dense $n \times n$ symmetric matrix involves $n(n+1)/2$ scalar variables and can become computationally expensive as n grows. To reduce this cost, Burer and Monteiro [BM03] proposed the low-rank factorization $Z = VV^\top$, with $V \in \mathbb{R}^{n\times m}$ and $m \ll n$. This reduces the number of variables to nm and enforces positive semidefiniteness automatically. The resulting problem is nonconvex, however, raising the possibility of nonglobal local minima. A central question is therefore how large m must be to guarantee that every second-order stationary point of the factorized problem is globally optimal.

Boumal, Voroninski, and Bandeira [BVB20] established such guarantees for smooth equality-constrained semidefinite programs. In the unit-diagonal setting, their generic-cost guarantee applies when $m(m+1)/2 > n$, while their uniform guarantee for all cost matrices applies when $m > n/2$ [BVB20, Theorem 1.4 and Corollary 5.11]. Waldspurger and Waters [WW20] showed that, in general, this generic rank threshold must grow on the order of the square root of the number of constraints, even for problems with a rank-one global solution. A bound independent of n therefore calls for additional structure.

The bounds above depend on the matrix size n. For matrices of rank-width at most k, Theorem 1.1 gives the sufficient bound $m \geq 2k$, independent of n. We impose no genericity assumption on the cost matrix A, but require the second-order stationary point V^* to be non-flat.

Our argument starts from the rank-width-one synchronization mechanism in Wu [Wu]. For nonnegative weights and factorization rank $m = 2$, that work proves that every local minimum is synchronized when the coupling matrix has connected support and rank-width one over $\mathbb{R}$. The proof merges synchronized vertex sets along a rank-decomposition tree. Since synchronized configurations are globally optimal in that setting, proving synchronization establishes global optimality.

Signed weights introduce a different difficulty: a global minimizer need not be synchronized, and its configuration is not known in advance. We therefore compare the energy at the stationary configuration with the energy at an arbitrary feasible configuration. We split this difference into the changes obtained by moving each side of a cut separately, minus a cross term. We prove lower bounds for the two single-set changes by induction on a rank-decomposition tree. Together, these bounds are large enough to offset the cross term, so the total energy cannot decrease.

1.3. **Organization.** Section 2 introduces the first- and second-order conditions and non-flatness. We then define rank-width over $\mathbb{R}$ and show how a cut factorization expresses the interaction between two vertex sets in terms of their aggregate vectors. Section 3 establishes the energy bound used in the proof. The key step is an induction argument: bounds for the energy changes on two child sets are combined to obtain the corresponding bound on their union. We first give this argument for rank-width one, then extend it to rank-width $k \geq 2$. Section 4 applies the bounds to the two sides of the root cut. Together, they control the cross term in the total energy change and show that no feasible configuration has lower energy than V^*, completing the proof of Theorem 1.1.

2. Preliminaries

2.1. **Notation. Sets and matrices.** Write $[n] = \{1, \dots, n\}$ and $S^c = [n] \setminus S$. For index sets S, T, $A[S, T]$ denotes the corresponding submatrix, and V_S consists of the rows of V indexed by S. The notation $L \sqcup R$ means a disjoint union. All matrix ranks are taken over $\mathbb{R}$. For a symmetric matrix, $C \succeq 0$ and $C \succ 0$ mean positive semidefiniteness and positive definiteness, respectively.

Vectors. We use $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ for the Euclidean inner product and norm, and $\|\cdot\|_F$ for the Frobenius norm. Consider a change from a fixed original configuration V^* to another feasible configuration V. At each vertex i, the vector changes from v_i^* to v_i; we denote its change by $\Delta v_i = v_i - v_i^*$.

2.2. **Stationary points and non-flat directions.** We call a feasible configuration V^* a *second-order stationary point* (SOSP) if

$$\operatorname{grad} E(V^*) = 0, \qquad \operatorname{Hess} E(V^*) \succeq 0,$$

where grad and Hess are the Riemannian gradient and Hessian on $(\mathbb{S}^{m-1})^n$, and positive semidefiniteness is understood on the tangent space at V^*. We also impose the following condition on motions that leave the energy unchanged.

Definition 2.1 (Non-flatness)**.** *A configuration V^* is* non-flat *if the following holds for every choice of fixed skew-symmetric matrices $\Omega_i \in \mathbb{R}^{m \times m}$. If the energy remains constant along*

$$v_i(t) = e^{t\Omega_i} v_i^*$$

for all sufficiently small t, then every inner product $\langle v_i(t), v_j(t) \rangle$ also remains constant for all sufficiently small t.

Thus a motion of this form can preserve the energy only if it preserves all pairwise inner products. Rotating all vectors together is allowed. When $m = 2$, write $\Omega_i = \omega_i J$, where J is the generator of planar rotations. The paths become $\theta_i(t) = \theta_i^* + t\omega_i$. Their full Gram matrix is constant precisely when all ω_i agree. Thus the definition agrees with non-flatness along straight phase directions in the planar problem. The condition concerns paths along which the energy is exactly constant; the Hessian may still have additional zero directions.

Lemma 2.2 (Stationarity conditions)**.** *Define*

$$\mu_i = \sum_j A_{ij}\langle v_i^*, v_j^*\rangle, \qquad M = \operatorname{diag}(\mu) - A. \tag{3}$$

At an SOSP,

$$MV^* = 0, \qquad \operatorname{tr}(\Xi^\top M \Xi) \geq 0 \quad \textit{whenever } \langle \xi_i, v_i^*\rangle = 0 \textit{ for every } i. \tag{4}$$

Proof. First-order stationarity on each sphere gives $\sum_j A_{ij} v_j^* = \mu_i v_i^*$, hence $MV^* = 0$. For a feasible curve, write $\xi_i = v_i'(0)$ and $a_i = v_i''(0)$. Differentiating $\|v_i(t)\|^2 = 1$ twice gives $\langle a_i, v_i^*\rangle = -\|\xi_i\|^2$. Substituting this into the second derivative of the energy at a stationary point gives

$$E''(0) = \sum_i \mu_i \|\xi_i\|^2 - \sum_{i,j} A_{ij}\langle \xi_i, \xi_j\rangle = \operatorname{tr}(\Xi^\top M \Xi).$$

Every tangent direction is the velocity of a feasible curve. The second-order condition therefore gives (4) for every matrix Ξ whose rows satisfy $\xi_i \perp v_i^*$.

□

Lemma 2.3 (Nonzero cuts and positive single-vertex coefficients)**.** *Suppose $n \geq 2$ and V^* is a non-flat SOSP with $m \geq 2$. Every nontrivial cut of A is nonzero, and $\mu_i > 0$ for every node i.*

Proof. If $A[S, S^c] = 0$ for a nontrivial cut, rotating all vectors in S together preserves the energy: the internal inner products stay fixed, and there is no interaction across the cut. Such a rotation can still change an inner product between a node in S and one in S^c: for any two nonzero vectors in dimension at least two, a suitable plane rotation of the first makes their inner product nonconstant. This contradicts non-flatness.

Next, choose a nonzero tangent direction that moves only node i. The second-order condition (4) gives $\mu_i \geq 0$. If $\mu_i = 0$, first-order stationarity gives $\sum_j A_{ij} v_j^* = 0$. Every rotation of node i alone then preserves the energy. A suitable rotation changes an inner product with another node, again a contradiction. □

2.3. **Rank-width.** For $S \subseteq [n] = \{1, \ldots, n\}$, let

$$\rho_A(S) = \operatorname{rank}_{\mathbb{R}} A[S, S^c].$$

A rank-decomposition is a tree of maximum degree three with one leaf for each vertex in $[n]$. Removing an edge separates the leaves into two sets S and S^c. The width of that edge is $\rho_A(S)$. The *rank-width over* $\mathbb{R}$ $\operatorname{rw}_{\mathbb{R}}(A)$ is the minimum, over such trees, of the maximum edge width. We set $\operatorname{rw}_{\mathbb{R}}(A) = 0$ for $n \leq 1$. A degree-two internal vertex can be removed by replacing its two incident edges with a single edge; this does not change the cuts represented by the tree. We use the matrix rank-width framework of Oum [Oum12], based on the tree-and-cut construction of Oum and Seymour [OS06]. Throughout this paper, the field is $\mathbb{R}$. For an unweighted graph, this need not agree with rank-width computed over $\mathbb{F}_2$.

To root the decomposition, first remove all degree-two internal vertices. Then insert a new vertex on one edge and use it as the root. Each internal vertex now has two children. For a vertex other than the root, let S be the set of leaves below it. The edge joining this vertex to its parent separates S from S^c. The induction proves a bound for each such set S, starting with individual leaves and then joining the sets below two children.

2.4. **Aggregate vectors across a cut.** Let $S \subseteq [n]$ and suppose the cut matrix $A[S, S^c]$ has rank $r_S \geq 1$. We will express the energy across this cut using r_S weighted sums of vectors on each side. Choose a factorization

$$A[S, S^c] = P_S Q_S^\top, \qquad P_S \in \mathbb{R}^{|S| \times r_S}, \quad Q_S \in \mathbb{R}^{|S^c| \times r_S}. \tag{5}$$

Both factors have full column rank. For $a = 1, \dots, r_S$, the ath column of P_S defines an aggregate vector on S. Collect these vectors as the columns of X_S:

$$x_S^a = \sum_{i \in S} (P_S)_{ia} v_i, \qquad X_S = V_S^\top P_S = [x_S^1 \ \cdots \ x_S^{r_S}] \in \mathbb{R}^{m \times r_S}. \tag{6}$$

When only S moves, the vectors in S^c remain at their original values. Using the columns of Q_S as weights, define their aggregates by

$$(w_S^a)^* = \sum_{j \in S^c} (Q_S)_{ja} v_j^*, \qquad a = 1, \dots, r_S,$$
$$W_S^* = [(w_S^1)^* \ \cdots \ (w_S^{r_S})^*] = (V_{S^c}^*)^\top Q_S.$$

Substituting the factorization into the crossing energy gives

$$-\sum_{i \in S, j \in S^c} A_{ij} \langle v_i, v_j^* \rangle = -\sum_{a=1}^{r_S} \langle x_S^a, (w_S^a)^* \rangle = -\operatorname{tr}(X_S (W_S^*)^\top). \tag{7}$$

Thus the crossing energy is a sum of r_S inner products between aggregate vectors. In particular, a rank-one cut requires one aggregate vector on each side. The weights in P_S and Q_S may have either sign.

We keep these factors fixed when comparing configurations. The change in the aggregates is therefore

$$\Delta X_S = X_S - X_S^* = (\Delta V_S)^\top P_S.$$

When two sets L and R are joined to form S, we need to relate their aggregate vectors. The next lemma expresses the aggregates of S and the interaction between L and R using the aggregates of the two children.

Lemma 2.4 (Relating the aggregates of a set and its children)**.** *Let S be the leaf set below a nonroot vertex, with children whose leaf sets are L and R, so $S = L \sqcup R$. Choose full-column-rank cut factorizations for L, R, S. There are matrices*

$$\Gamma \in \mathbb{R}^{r_L \times r_R}, \qquad H \in \mathbb{R}^{(r_L + r_R) \times r_S}$$

such that, with $P_0 = \operatorname{diag}(P_L, P_R)$,

$$A[L, R] = P_L \Gamma P_R^\top, \qquad P_S = P_0 H. \tag{8}$$

The matrix H has full column rank, and, writing $X_0 = [X_L \ X_R]$,

$$X_S = X_0 H. \tag{9}$$

Proof. The matrix $A[L, R]$ is part of the cut matrix for L. Its columns therefore lie in range P_L. Applying the same argument to the cut for R shows that its rows lie in the row space of $P_R^\top$. These two inclusions give $A[L, R] = P_L \Gamma P_R^\top$ for some Γ. Similarly, the rows of P_S indexed by L can be written as P_L times a coefficient matrix, and the rows indexed by R can be written using P_R. Stacking these two coefficient matrices gives H. The full column rank of $P_S = P_0 H$ implies the full column rank of H. Multiplication by $V_S^\top$ proves (9). □

Lemma 2.5 (Alignment across a rank-one cut)**.** *Assume $m \geq 2$. At a non-flat SOSP, let $A[S, S^c] = pq^\top$ be a nonzero rank-one cut. Put $u^* = \sum_{i \in S} p_i v_i^*$ and $w^* = \sum_{j \notin S} q_j v_j^*$. Then*

$$u^* \neq 0, \qquad w^* \neq 0, \qquad w^* = \lambda_S u^*, \quad \lambda_S > 0. \tag{10}$$

Proof. If either aggregate were zero, rotating all vectors in S together would preserve the crossing energy $-\langle u^*, w^* \rangle$ and all internal energies. We could choose a rotation that changes an inner product between a vector in S and one in S^c, violating non-flatness. Thus both aggregates are nonzero.

Next, rotate the vectors in S by $e^{t\Omega}$ while fixing S^c. The first-order condition gives $\langle \Omega u^*, w^* \rangle = 0$. If w^* had a nonzero component $a \perp u^*$, choose $\Omega = (a(u^*)^\top - u^* a^\top)/\|u^*\|^2$ so that $\Omega u^* = a$. The first-order condition would then give $\|a\|^2 = 0$, a contradiction. Thus $w^* = \lambda_S u^*$ for some scalar λ_S. To determine its sign, rotate S at unit angular speed in a plane containing u^*. The second derivative of the crossing energy is $\lambda_S \|u^*\|^2$, which must be nonnegative. Since both aggregates are nonzero, $\lambda_S > 0$. □

3. Splitting and lower bounding energy changes

Fix a non-flat second-order stationary point V^* and an arbitrary feasible configuration V. We will prove that $E(V) \geq E(V^*)$.

3.1. **The split of energy changes.** We first compare moving two disjoint sets L and R together with moving each set separately from the same original configuration V^*. The difference is a cross term involving the changes on both sides. Taking $L \cup R = [n]$ then gives a decomposition of the total energy change from V^* to V.

To better present this decomposition, we start with some notation. The matrix $V \in \mathbb{R}^{n\times m}$ records the vectors at the n vertices: its ith row is $v_i^\top$, where $v_i \in \mathbb{R}^m$ and $\|v_i\| = 1$. For $S \subseteq [n]$, let V_S consist of the rows indexed by S. The total energy of this configuration is

$$E(V) = -\sum_{1\leq i<j\leq n} A_{ij}\langle v_i, v_j\rangle = -\frac{1}{2}\operatorname{tr}(V^\top A V).$$

Define the energy of the edges within S by

$$E_S(V_S) = -\sum_{\substack{i,j\in S\\ i<j}} A_{ij}\langle v_i, v_j\rangle,$$

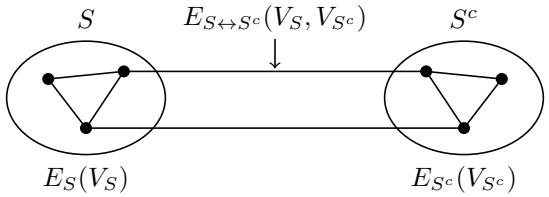
and the energy of the edges joining S to S^c by

$$E_{S\leftrightarrow S^c}(V_S, V_{S^c}) = -\sum_{i\in S,\, j\in S^c} A_{ij}\langle v_i, v_j\rangle. \tag{11}$$

The total energy is therefore

$$E(V) = E_S(V_S) + E_{S^c}(V_{S^c}) + E_{S\leftrightarrow S^c}(V_S, V_{S^c}). \tag{12}$$

The following figure illustrates how the cut (S, S^c) decomposes $E(V)$ into these three terms.



Fix an original configuration V^* and a new feasible configuration V, and write $\Delta v_i = v_i - v_i^*$. Moving only S means replacing v_i^* by v_i for $i \in S$, while keeping all vectors in S^c at their original values. We denote the total energy of this configuration by $E(V_S, V^*_{S^c})$ and define

$$\begin{aligned}\Delta E_S &= E(V_S, V^*_{S^c}) - E(V^*_S, V^*_{S^c})\\ &= \left[E_S(V_S) - E_S(V^*_S)\right]\\ &\quad + \left[E_{S\leftrightarrow S^c}(V_S, V^*_{S^c}) - E_{S\leftrightarrow S^c}(V^*_S, V^*_{S^c})\right].\end{aligned} \tag{13}$$

The energy of the edges entirely within S^c remains unchanged and cancels in the difference. Thus, ΔE_S consists of the internal energy change within S and the energy change across the cut. In particular, $\Delta E_{[n]} = E(V) - E(V^*)$.

Lemma 3.1. *For disjoint vertex sets $L, R \subseteq [n]$, one has*

$$\Delta E_{L\cup R} = \Delta E_L + \Delta E_R - \sum_{i\in L,\, j\in R} A_{ij}\langle \Delta v_i, \Delta v_j\rangle. \tag{14}$$

Proof. Starting from V^*, first move the vectors in L to their new values, and then move those in R. The first move changes the energy by ΔE_L, so the second changes it by $\Delta E_{L\cup R} - \Delta E_L$.

Compare this second change with ΔE_R, which is obtained by moving R directly from V^*. The only difference is that the vectors in L are now v_i rather than v_i^*. Edges within R and edges joining R to $[n]\setminus(L\cup R)$ therefore have the same energy changes in both cases. For an edge joining $i \in L$ to $j \in R$, the change is $-A_{ij}\langle v_i, \Delta v_j\rangle$ after L has moved, compared with $-A_{ij}\langle v_i^*, \Delta v_j\rangle$ when starting from V^*. Their difference is

$$-A_{ij}\langle v_i, \Delta v_j\rangle + A_{ij}\langle v_i^*, \Delta v_j\rangle = -A_{ij}\langle \Delta v_i, \Delta v_j\rangle.$$

Summing over the edges between L and R gives

$$\Delta E_{L\cup R} - \Delta E_L - \Delta E_R = -\sum_{i\in L,\, j\in R} A_{ij}\langle \Delta v_i, \Delta v_j\rangle,$$

which rearranges to (14). □

3.2. **Lower bounding ΔE_S.**

Lemma 3.2 (Lower bound when only a subtree moves)**.** *Let V^* be a non-flat second-order stationary point under the hypotheses of Theorem 1.1. We can choose a rank-decomposition of width at most k and a root with the following property. Let S be the set of leaves below any nonroot vertex, and choose any full-column-rank factorization (5). Then there is a fixed symmetric positive definite matrix $T_S \in \mathbb{R}^{r_S\times r_S}$ such that*

$$\Delta E_S \geq \frac{1}{2}\operatorname{tr}(\Delta X_S T_S \Delta X_S^\top) \qquad \textit{for every feasible new configuration.} \tag{15}$$

3.2.1. *Rank-width one.*

Proof of Lemma 3.2 for $k = 1$. If $n = 1$, there are no nonroot subtrees, so there is nothing to prove. Hence assume $n \geq 2$. Since $\operatorname{rw}_{\mathbb{R}}(A) \leq 1$, there exists a rank-decomposition tree in which every edge induces a cut matrix of rank at most one. Every such cut is nonzero by Lemma 2.3, and hence has rank one. Thus $r_S = 1$, X_S consists of a single aggregate vector u_S, and (15) reduces to

$$\Delta E_S \geq \frac{\lambda_S}{2}\|\Delta u_S\|^2, \qquad T_S = [\lambda_S].$$

Remove degree-two internal vertices as described in Section 2.3. Insert the root on any edge of the tree. Each nonroot vertex represents the set S of leaves below it, and each internal vertex has two children. For each such S, write $A[S, S^c] = p_S q_S^\top$ and set

$$u_S = \sum_{i\in S}(p_S)_i v_i, \qquad w_S^* = \sum_{j\in S^c}(q_S)_j v_j^*.$$

Lemma 2.5 gives a fixed positive coefficient λ_S satisfying $w_S^* = \lambda_S u_S^*$. We prove the scalar bound above with these coefficients.

Base case. Let $S = \{i\}$ be the vertex set corresponding to a leaf. The original aggregate vector is $u_S^* = (p_S)_i v_i^*$, and the new aggregate vector is $u_S = (p_S)_i v_i$. Since v_i^* and v_i are unit vectors and the weight $(p_S)_i$ is fixed,

$$\|u_S^*\| = \|u_S\| = |(p_S)_i|.$$

There are no edges within S, so only the energy across the cut changes. Using $w_S^* = \lambda_S u_S^*$, we obtain

$$\Delta E_S = -\langle \Delta u_S, w_S^*\rangle = -\lambda_S\langle \Delta u_S, u_S^*\rangle = \frac{\lambda_S}{2}\|\Delta u_S\|^2.$$

The last equality follows by expanding $\|\Delta u_S\|^2 = \|u_S - u_S^*\|^2$ and using $\|u_S\| = \|u_S^*\|$. Thus the desired bound holds with equality for a single vertex.

Induction hypothesis. Let $S = L \sqcup R$ and assume the scalar bounds hold for both children with their alignment coefficients λ_L, λ_R.

Inductive step. Lemma 2.4 gives scalar coefficients c, d, γ such that $u_S = cu_L + du_R$ and $A[L, R] = \gamma p_L p_R^\top$. Using Lemma 3.1 and the two child bounds, we obtain

$$\begin{aligned}\Delta E_S &= \Delta E_L + \Delta E_R - \gamma\langle \Delta u_L, \Delta u_R\rangle \\ &\geq \frac{\lambda_L}{2}\|\Delta u_L\|^2 + \frac{\lambda_R}{2}\|\Delta u_R\|^2 - \gamma\langle \Delta u_L, \Delta u_R\rangle \\ &= \frac{\lambda_S}{2}\|\Delta u_S\|^2 + Q.\end{aligned}$$

The two child bounds give the desired bound for S plus a remainder Q. It is therefore enough to prove $Q \geq 0$. Expanding $\|\Delta u_S\|^2 = \|c\Delta u_L + d\Delta u_R\|^2$ gives

$$Q = \frac{a}{2}\|\Delta u_L\|^2 + \frac{b}{2}\|\Delta u_R\|^2 - \kappa\langle \Delta u_L, \Delta u_R\rangle, \tag{16}$$

where

$$a = \lambda_L - \lambda_S c^2, \quad b = \lambda_R - \lambda_S d^2, \quad \kappa = \gamma + \lambda_S cd.$$

The cut factorizations in Lemma 2.4 give $w_L^* = \gamma u_R^* + cw_S^*$ and $w_R^* = \gamma u_L^* + dw_S^*$. Applying Lemma 2.5 to the three cuts yields

$$\gamma u_R^* + cw_S^* = \lambda_L u_L^*, \quad \gamma u_L^* + dw_S^* = \lambda_R u_R^*, \quad w_S^* = \lambda_S(cu_L^* + du_R^*).$$

Substitute the third relation into the first two and collect the coefficients of u_L^* and u_R^*. This gives

$$au_L^* = \kappa u_R^*, \qquad bu_R^* = \kappa u_L^*. \tag{17}$$

If u_L^*, u_R^* are linearly independent, the first equation forces $a = \kappa = 0$, and the second then gives $b = 0$. Thus $a = b = \kappa = 0$, so $Q = 0$.

If they are parallel, write $u_L^* = \rho e$, $u_R^* = \sigma e$, where $\|e\| = 1$ and $\rho\sigma \neq 0$; ρ, σ may have either sign. Set $\eta = \kappa/(\rho\sigma)$. Equation (17) gives $a = \eta\sigma^2$, $b = \eta\rho^2$, and therefore

$$Q = \frac{\eta}{2}\|\sigma\Delta u_L - \rho\Delta u_R\|^2. \tag{18}$$

It remains to show that the coefficient η is nonnegative. Choose a two-dimensional plane containing e, and let $\mathcal{R}_t$ rotate through angle t in this plane while fixing its orthogonal complement. Apply $\mathcal{R}_{\alpha t}$ to every vector in L and $\mathcal{R}_{\beta t}$ to every vector in R, keeping the exterior fixed. The node vectors need not all lie in the selected plane; their aggregate vectors do. Internal energies of the two children remain unchanged. The varying energy is

$$-\gamma\langle \mathcal{R}_{\alpha t}u_L^*, \mathcal{R}_{\beta t}u_R^*\rangle - c\langle \mathcal{R}_{\alpha t}u_L^*, w_S^*\rangle - d\langle \mathcal{R}_{\beta t}u_R^*, w_S^*\rangle.$$

At $t = 0$, the second-order condition and the alignment equations give

$$0 \leq \lambda_L\rho^2\alpha^2 + \lambda_R\sigma^2\beta^2 - 2\gamma\rho\sigma\alpha\beta.$$

Choose $\alpha = d/\rho$ and $\beta = -c/\sigma$ to cancel the terms containing λ_S. Then

$$\begin{aligned}0 &\leq \lambda_L d^2 + \lambda_R c^2 + 2\gamma cd \\ &= ad^2 + bc^2 + 2\kappa cd = \eta(c\rho + d\sigma)^2.\end{aligned}$$

By Lemma 2.5, the parent aggregate $u_S^* = (c\rho + d\sigma)e$ is nonzero. Hence $\eta \geq 0$ and (18) gives $Q \geq 0$, proving the parent bound. This completes the induction for $k = 1$.

□

3.2.2. *Rank-width $k \geq 2$.* We first establish two geometric facts used in the induction. The merge argument below uses directions perpendicular to the original vectors in the set being merged. The first lemma bounds the dimension spanned by the vectors on one side of a cut. The second uses this bound to choose a root so that these perpendicular directions are available at all but one nonroot merge when the original configuration spans $\mathbb{R}^m$. Write $U_S = \operatorname{span}\{v_i^* : i \in S\}$ for the space spanned by the original vectors in S. Its orthogonal complement $U_S^\perp$ consists of the vectors perpendicular to every v_i^* with $i \in S$.

Lemma 3.3 (The vectors in a set span the same space as its external forces)**.** *Let V^* be a non-flat stationary point. Suppose $\rho_A(S) \leq k$, $m \geq k+2$, and $\dim U_{S^c} > k$. For each $i \in S$, let h_i be the force from the vectors outside S, and let W be the space spanned by these forces:*

$$h_i = \sum_{j \notin S} A_{ij} v_j^*, \qquad W = \operatorname{span}\{h_i : i \in S\},$$

Then

$$U_S = W \subseteq U_{S^c}, \qquad \dim U_S \leq k. \tag{19}$$

Proof. The matrix whose rows are $h_i^\top$ is $A[S, S^c]V_{S^c}^*$. Its rank is at most k, so $\dim W \leq k$. Each h_i is a linear combination of exterior vectors, hence $W \subseteq U_{S^c}$.

We first show that every vector in S belongs to W. Suppose instead that some v_i^* has a nonzero projection a onto $W^\perp$. Since $\dim U_{S^c} > k \geq \dim W$, some exterior vector also has a nonzero projection b onto $W^\perp$. Moreover, $m \geq k+2$ gives $\dim W^\perp \geq 2$, leaving room to rotate in this space. There is a skew-symmetric operator Ω on $W^\perp$ such that $\langle e^{t\Omega}a, b\rangle$ is nonconstant. If a, b are independent, take $\Omega = ba^\top - ab^\top$, for which

$$\langle \Omega a, b\rangle = \|a\|^2\|b\|^2 - \langle a, b\rangle^2 > 0.$$

If $b = ca$ with $c \neq 0$, rotate a plane containing a inside $W^\perp$; the inner product becomes $c\|a\|^2 \cos t$ after a choice of speed.

Extend Ω by zero on W and rotate all nodes in S with $e^{t\Omega}$, fixing its exterior. Internal energies are unchanged, and every crossing contribution $-\langle e^{t\Omega}v_i^*, h_i\rangle$ is unchanged because $h_i \in W$. However, the inner product between the selected interior and exterior vectors changes. This contradicts non-flatness, proving $U_S \subseteq W$. For the reverse inclusion, first-order stationarity gives

$$h_i = \mu_i v_i^* - \sum_{j \in S} A_{ij} v_j^* \in U_S.$$

This puts every h_i in U_S, so $W \subseteq U_S$. Combining the two inclusions proves (19). □

Lemma 3.4 (Choosing the root when the original vectors span $\mathbb{R}^m$)**.** *Suppose $k \geq 2$, $m \geq 2k$, $\operatorname{rw}_{\mathbb{R}}(A) \leq k$, and V^* is a non-flat stationary point with $\operatorname{rank} V^* = m$. Then $m = 2k$. Take any decomposition of width at most k and remove its degree-two internal vertices. There is an internal vertex whose three branches have leaf sets S_1, S_2, S_3 satisfy*

$$\dim U_{S_i} = k, \qquad U_{S_i} \oplus U_{S_j} = \mathbb{R}^{2k} \quad (i \neq j). \tag{20}$$

Place the root on the edge separating S_3 from $S_1 \cup S_2$. Every nonroot merge then takes place in a set S with $\dim U_S^\perp \geq 2$, except for $S_1, S_2 \longrightarrow S_1 \cup S_2$.

Proof. We first show that every tree-edge cut has exactly one side whose original vectors span more than k dimensions. Since $m \geq 2k \geq k+2$, Lemma 3.3 rules out the possibility that both sides span more than k dimensions.

Suppose both sides span at most k dimensions. If $m > 2k$, their sum cannot be all of $\mathbb{R}^m$, contrary to $\operatorname{rank} V^* = m$. If $m = 2k$, both spaces must have dimension k and their intersection must be zero. For $i \in S$, the force h_i belongs to U_{S^c} by definition and to U_S by first-order stationarity.

Hence $h_i = 0$. Rotating S together would then leave its energy unchanged, although a suitable rotation changes an inner product across the cut. This contradicts non-flatness.

Orient each edge toward its side of dimension greater than k. Following the arrows in a finite tree must eventually lead to a vertex with no outgoing edge. This vertex cannot be a leaf, since a single vector spans only one dimension. It is therefore an internal vertex with three incident edges, all pointing toward it. Each of its three branches lies on the smaller side of its edge, so the original vectors in each branch span at most k dimensions. For each branch, Lemma 3.3 gives

$$U_{S_i} \subseteq U_{S_j} + U_{S_\ell}.$$

Thus the vectors in any one branch lie in the space spanned by the other two. Any two branches therefore span the entire original configuration, which is $\mathbb{R}^m$. Each branch has dimension at most k, so $m \le 2k$. Together with $m \ge 2k$, this gives $m = 2k$. Each branch must have dimension k, and the spaces of any two branches have zero intersection. This proves (20).

Insert the root on the edge between S_3 and $S_1 \cup S_2$. Any subtree inside one branch has $\dim U_S \le k$, so $\dim U_S^{\perp} \ge m - k = k \ge 2$. Among the nonroot subtrees, the only one that contains more than one branch is $S_1 \cup S_2$. This is the only merge that must be treated separately. □

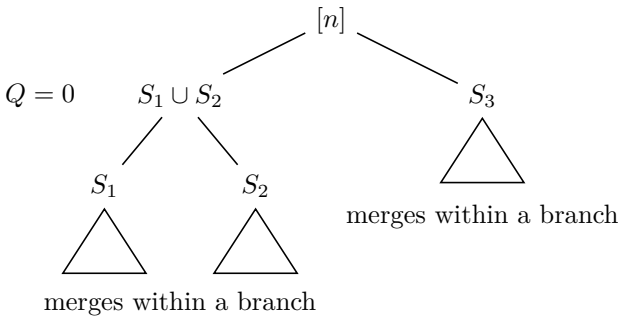


FIGURE 1. The root is placed on the edge separating S_3 from $S_1 \cup S_2$ so that the induction proceeds within each of the three branches first. Within each branch, the original vectors span at most k dimensions, so every merge has at least two perpendicular directions available. The only nonroot merge outside these branches is $S_1, S_2 \to S_1 \cup S_2$; it is treated separately below, where we prove $Q = 0$.

Proof of Lemma 3.2 for $k \ge 2$. If $n = 1$, there are no nonroot subtrees, so there is nothing to prove. Hence assume $n \ge 2$. Since $\mathrm{rw}_{\mathbb{R}}(A) \le k$, there exists a rank-decomposition tree in which every edge induces a cut matrix of rank at most k. Remove degree-two internal vertices as described in Section 2.3. If $\operatorname{rank} V^* < m$, choose any edge of this tree; if $\operatorname{rank} V^* = m$, choose the edge specified by Lemma 3.4. Insert a new vertex on the chosen edge and use it as the root. We first prove the bound for each leaf, then use the bounds for two children to prove the bound for their union.

Base case. We first prove the bound for a single vertex $S = \{i\}$. First-order stationarity gives $\sum_j A_{ij} v_j^* = \mu_i v_i^*$, with $\mu_i > 0$ by Lemma 2.3. Moving only this vertex therefore gives

$$\begin{aligned} \Delta E_{\{i\}} &= -\left\langle v_i - v_i^*, \sum_j A_{ij} v_j^* \right\rangle \\ &= -\mu_i \langle v_i - v_i^*, v_i^* \rangle = \frac{\mu_i}{2} \|v_i - v_i^*\|^2. \end{aligned}$$

The last equality uses $\|v_i\| = \|v_i^*\| = 1$. The cut factor is a nonzero scalar, $P_{\{i\}} = [p_i]$, so $\Delta X_{\{i\}} = p_i \Delta v_i$. Taking $T_{\{i\}} = [\mu_i / p_i^2] \succ 0$ gives

$$\Delta E_{\{i\}} = \frac{1}{2} \operatorname{tr}(\Delta X_{\{i\}} T_{\{i\}} \Delta X_{\{i\}}^\top),$$

which establishes the base case with equality.

Induction hypothesis. Let $S = L \sqcup R$, where L and R are the leaf sets of its two children. Assume that there exist fixed symmetric positive definite matrices T_L and T_R such that

$$\Delta E_L \geq \frac{1}{2} \operatorname{tr}(\Delta X_L T_L \Delta X_L^\top), \qquad \Delta E_R \geq \frac{1}{2} \operatorname{tr}(\Delta X_R T_R \Delta X_R^\top)$$

for every feasible new configuration.

Inductive step. Lemma 3.1 gives

$$\begin{aligned} \Delta E_S &= \Delta E_L + \Delta E_R - \sum_{i \in L,\, j \in R} A_{ij} \langle \Delta v_i, \Delta v_j \rangle \\ &\geq \frac{1}{2} \operatorname{tr}(\Delta X_L T_L \Delta X_L^\top) + \frac{1}{2} \operatorname{tr}(\Delta X_R T_R \Delta X_R^\top) - \sum_{i \in L,\, j \in R} A_{ij} \langle \Delta v_i, \Delta v_j \rangle. \end{aligned}$$

We want this right-hand side to be at least $\frac{1}{2} \operatorname{tr}(\Delta X_S T_S \Delta X_S^\top)$ for some fixed $T_S \succ 0$. Define the difference by

$$\begin{aligned} Q := &\frac{1}{2} \operatorname{tr}(\Delta X_L T_L \Delta X_L^\top) + \frac{1}{2} \operatorname{tr}(\Delta X_R T_R \Delta X_R^\top) \\ &- \sum_{i \in L,\, j \in R} A_{ij} \langle \Delta v_i, \Delta v_j \rangle - \frac{1}{2} \operatorname{tr}(\Delta X_S T_S \Delta X_S^\top). \end{aligned} \tag{21}$$

Thus

$$\Delta E_S \geq \frac{1}{2} \operatorname{tr}(\Delta X_S T_S \Delta X_S^\top) + Q.$$

To obtain (15), our goal boils down to finding a fixed $T_S \succ 0$ for which $Q \geq 0$ for every feasible new configuration.

Steps 1–3 establish the bound for a merge satisfying at least one of the following conditions:

(1) a nonzero vector is perpendicular to every original node vector;
(2) $\dim U_S^\perp \geq 2$.

Both conditions give directions perpendicular to the original vectors in S. We use them to test the second-order condition. Step 4 applies the resulting bound throughout the tree, and Step 5 treats the one remaining merge when the original vectors span $\mathbb{R}^m$.

Step 1. We construct child coefficients satisfying (24). The induction hypothesis gives at least one positive definite coefficient for each child. We will choose new coefficients that still satisfy the child bounds. We also prove the matrix identities in (24); these will let us apply the second-order condition to the remainder Q. For a child $I \in \{L, R\}$, denote the coefficient from the induction hypothesis by $T_I^{(0)}$, and write $M_I = M[I, I]$.

We first express the exact energy change as a quadratic form in ΔV_I. Let D equal ΔV_I on the rows in I and zero on all other rows, so $V^* + D$ is the configuration obtained by moving only I. Expanding the energy gives

$$E(V^* + D) - E(V^*) = -\operatorname{tr}(D^\top A V^*) - \tfrac{1}{2} \operatorname{tr}(D^\top A D).$$

Write the ith row of D as $d_i^\top$. Since both v_i^* and $v_i^* + d_i$ have unit length, $\langle d_i, v_i^* \rangle = -\|d_i\|^2/2$. Using $AV^* = \operatorname{diag}(\mu) V^*$ in the linear term gives

$$\Delta E_I = \frac{1}{2} \operatorname{tr}(\Delta V_I^\top M_I \Delta V_I). \tag{22}$$

We next use small movements to turn the child bound into a matrix inequality. Either condition (1) or (2) gives a unit vector e perpendicular to every original vector in I. For any coefficients $z \in \mathbb{R}^{|I|}$, move the vectors along

$$v_i(t) = \cos(tz_i)v_i^* + \sin(tz_i)e, \qquad i \in I,$$

while fixing the exterior. Each vector stays on the unit sphere, and its initial velocity is $z_i e$. Hence $\Delta V_I(t)/t \longrightarrow ze^\top$. The child bound holds along this curve. Substitute (22) into that bound, divide by t^2, and let $t \to 0$ to obtain

$$z^\top (M_I - P_I T_I^{(0)} P_I^\top) z \geq 0.$$

Since z was arbitrary, $M_I \succeq P_I T_I^{(0)} P_I^\top \succeq 0$. If $M_I z = 0$, this inequality and $T_I^{(0)} \succ 0$ force $P_I^\top z = 0$. Thus $\ker M_I \subseteq \ker P_I^\top$: every column of P_I is orthogonal to $\ker M_I$ and therefore lies in the range of M_I.

The matrix M_I may be singular, so we use its pseudoinverse $M_I^\dagger$. For a positive semidefinite matrix, this replaces each positive eigenvalue by its reciprocal and leaves each zero eigenvalue at zero. Define the new coefficient T_I and two auxiliary matrices by

$$\begin{aligned} T_I &= (P_I^\top M_I^\dagger P_I)^{-1}, \\ C_I &= M_I - P_I T_I P_I^\top, \qquad Z_I = M_I^\dagger P_I T_I. \end{aligned} \tag{23}$$

Here C_I is the part of M_I left after subtracting $P_I T_I P_I^\top$. We will prove that $C_I \succeq 0$, which ensures that the new coefficient still gives a lower bound. The role of Z_I is to produce perturbations for which this remaining part contributes zero.

All three matrices depend only on V^* and the fixed cut factors. To check that T_I exists and is positive definite, put $D_I = M_I^{\dagger/2} P_I$, where $M_I^{\dagger/2}$ is the positive semidefinite square root of $M_I^\dagger$. The columns of P_I are independent and belong to the range of M_I, so D_I also has full column rank. Therefore $T_I = (D_I^\top D_I)^{-1} \succ 0$. Moreover, $M_I^{1/2} D_I = P_I$, which gives

$$C_I = M_I^{1/2}[I - D_I(D_I^\top D_I)^{-1} D_I^\top] M_I^{1/2} \succeq 0,$$

The bracket is the orthogonal projection onto the space perpendicular to the columns of D_I, so it is positive semidefinite. Direct multiplication also gives $P_I^\top Z_I = I$ and $M_I Z_I = P_I T_I$, hence $C_I Z_I = 0$. In particular, for $z = Z_I x$ we have $P_I^\top z = x$ and $z^\top C_I z = 0$. These are the identities we will use to test the second-order condition in Step 2.

We also need $C_I V_I^* = 0$. The definition of C_I shows that it vanishes on $\ker M_I$ and satisfies $C_I M_I^\dagger P_I = 0$. First-order stationarity gives $M_I V_I^* = P_I (W_I^*)^\top$, so

$$C_I V_I^* = C_I M_I^\dagger M_I V_I^* = C_I M_I^\dagger P_I (W_I^*)^\top = 0.$$

We have now proved all of the following identities:

$$\begin{aligned} &M_I = C_I + P_I T_I P_I^\top, \qquad T_I \succ 0, \quad C_I \succeq 0, \\ &C_I V_I^* = 0, \qquad P_I^\top Z_I = I, \qquad C_I Z_I = 0. \end{aligned} \tag{24}$$

These identities follow from the child bounds. In particular, substituting $M_I = C_I + P_I T_I P_I^\top$ into (22) shows that the new coefficient gives the required bound for every feasible movement:

$$\Delta E_I = \frac{1}{2}\operatorname{tr}(\Delta X_I T_I \Delta X_I^\top) + \frac{1}{2}\operatorname{tr}(\Delta V_I^\top C_I \Delta V_I) \geq \frac{1}{2}\operatorname{tr}(\Delta X_I T_I \Delta X_I^\top). \tag{25}$$

Finally, substitute this decomposition of M_I into $M_I V_I^* = P_I (W_I^*)^\top$. Since $C_I V_I^* = 0$ and P_I has full column rank, we can cancel P_I and obtain

$$W_I^* = X_I^* T_I. \tag{26}$$

We can now combine the two child bounds. Use the new coefficients in (21), and take Γ, H from Lemma 2.4. Collect the two sets of aggregate vectors in $X_0 = [X_L\ X_R]$, and collect their coefficients in

$$J = \begin{pmatrix} T_L & -\Gamma \\ -\Gamma^\top & T_R \end{pmatrix}. \tag{27}$$

The crossing term is $\operatorname{tr}(\Delta X_L \Gamma \Delta X_R^\top)$, so the combined child bound is

$$\Delta E_S \geq \frac{1}{2} \operatorname{tr}(\Delta X_0 J \Delta X_0^\top). \tag{28}$$

Since $\Delta X_S = \Delta X_0 H$, the remainder in (21) has the form

$$\mathcal{Q}_S = J - H T_S H^\top, \qquad Q = \frac{1}{2} \operatorname{tr}(\Delta X_0 \mathcal{Q}_S \Delta X_0^\top). \tag{29}$$

Here $\mathcal{Q}_S$ is the coefficient matrix of the scalar remainder Q. If we can choose $T_S \succ 0$ so that $\mathcal{Q}_S \succeq 0$, then $Q \geq 0$ for every change in the aggregate vectors, as required.

Step 2. We show that $J \succeq 0$ *and* $\ker J \subseteq \ker H^\top$. We first prove $J \succeq 0$. Form the block diagonal matrices

$$C_0 = \operatorname{diag}(C_L, C_R), \qquad Z_0 = \operatorname{diag}(Z_L, Z_R), \qquad P_0 = \operatorname{diag}(P_L, P_R).$$

The diagonal blocks of $M[S,S]$ are M_L and M_R, and its off-diagonal block is $-A[L,R] = -P_L \Gamma P_R^\top$. Using (24) for the two children gives

$$M[S,S] = C_0 + P_0 J P_0^\top, \qquad P_0^\top Z_0 = I, \qquad C_0 Z_0 = 0, \qquad C_0 V_S^* = 0. \tag{30}$$

To test J in an arbitrary direction $x \in \mathbb{R}^{r_L + r_R}$, set $z = Z_0 x$ and choose a unit vector $e \in U_S^\perp$. Move each vector in S initially along e, with velocity $\xi_i = z_i e$ for $i \in S$ and $\xi_j = 0$ outside S. These velocities are tangent to the spheres because $e \perp v_i^*$ for every $i \in S$. Using $C_0 Z_0 = 0$ and $P_0^\top Z_0 = I$, the second-order condition becomes

$$0 \leq \operatorname{tr}(\Xi^\top M \Xi) = z^\top M[S,S] z = x^\top J x.$$

Thus $J \succeq 0$.

We next show that $Jx = 0$ implies $H^\top x = 0$. This will allow us to construct T_S even when J is singular. If $x \in \ker J$, the perturbation just constructed has zero second variation. The Hessian is positive semidefinite on the tangent space, so its mixed term between this perturbation and any other tangent direction must vanish. Indeed, nonnegativity on $\Xi + t\Upsilon$ for both signs of t rules out a nonzero linear term in t. Choose Υ to move only an exterior node j, with velocity $\zeta_j \perp v_j^*$. The mixed term gives

$$\left(\sum_{i \in S} A_{ji} z_i \right) \langle e, \zeta_j \rangle = 0. \tag{31}$$

Under condition (1), choose e perpendicular to all original vectors and take $\zeta_j = e$. The inner product is then one, so the sum in parentheses is zero. Under condition (2), we may choose any unit $e \in U_S^\perp$. If the sum were nonzero, the equation would force every such e to be orthogonal to all $\zeta_j \perp v_j^*$, and hence parallel to v_j^*. A space of dimension at least two cannot be contained in this single line. The sum must again be zero. Therefore $A[S^c, S] z = 0$ in either case. Since $P_S = P_0 H$,

$$0 = A[S^c, S] z = Q_S H^\top P_0^\top Z_0 x = Q_S H^\top x.$$

Since Q_S has full column rank, $Q_S H^\top x = 0$ implies $H^\top x = 0$. As $x \in \ker J$ was arbitrary, we obtain

$$\ker J \subseteq \ker H^\top. \tag{32}$$

Step 3. We construct $T_S \succ 0$ for which $Q \geq 0$. By (32), every column of H is orthogonal to $\ker J$ and therefore lies in the range of J. On that range, $J^\dagger$ is positive definite. Since H has full column rank, $H^\top J^\dagger H \succ 0$. Its inverse exists, so we choose

$$T_S = (H^\top J^\dagger H)^{-1} \succ 0. \tag{33}$$

We now check that this choice gives $Q \geq 0$. Put $D_S = J^{\dagger/2}H$. Then $J^{1/2}D_S = H$ and $D_S^\top D_S = T_S^{-1}$, so

$$\mathcal{Q}_S = J - HT_SH^\top = J^{1/2}[I - D_S(D_S^\top D_S)^{-1}D_S^\top]J^{1/2} \succeq 0. \tag{34}$$

The bracket projects onto the space perpendicular to the columns of D_S and is therefore positive semidefinite. Thus $\mathcal{Q}_S \succeq 0$, so (29) gives $Q \geq 0$ for every feasible movement. Substitution into the energy inequality proves the desired bound for S.

We also construct matrices C_S, Z_S satisfying the same identities as the child matrices in (24). These identities will be used when combining bounds at the root. Set $G = J^\dagger HT_S$. From the definition of T_S and the fact that H lies in the range of J, we obtain

$$H^\top G = I, \qquad JG = HT_S, \qquad \mathcal{Q}_S G = 0. \tag{35}$$

Define

$$C_S = C_0 + P_0\mathcal{Q}_S P_0^\top, \qquad Z_S = Z_0 G. \tag{36}$$

Then $C_S \succeq 0$, $M[S,S] = C_S + P_S T_S P_S^\top$, $P_S^\top Z_S = I$, and $C_S Z_S = 0$. It remains to check $C_S V_S^* = 0$. First-order stationarity reads $M[S,S]V_S^* = P_S(W_S^*)^\top$. Substitute (30), use $C_0 V_S^* = 0$ and $P_S = P_0 H$, and cancel the full-column-rank matrix P_0. The result is

$$J(X_0^*)^\top = H(W_S^*)^\top, \qquad X_0^* = [X_L^*\ X_R^*]. \tag{37}$$

Multiplying by $G^\top$ gives $(W_S^*)^\top = T_S(X_S^*)^\top$. Substituting this back yields $\mathcal{Q}_S(X_0^*)^\top = 0$, and hence $C_S V_S^* = 0$. This verifies (24) for S. The only induction assumptions used were the two child energy bounds.

Step 4. We prove the bounds along the rooted tree. If $\operatorname{rank} V^* < m$, a vector perpendicular to all original vectors exists, so condition (1) applies at every nonroot merge. Starting at the leaves, Steps 1–3 therefore prove the bound for every nonroot subtree.

Suppose instead that $\operatorname{rank} V^* = m$. By Lemma 3.4, condition (2) applies to every merge inside the three branches S_1, S_2, S_3. Each such subtree has vector span of dimension at most k, so its orthogonal complement has dimension at least $m-k \geq 2$. Starting at the leaves, Steps 1–3 therefore establish the bounds throughout all three branches. The only remaining nonroot set is $S_1 \cup S_2$.

Step 5. We prove the bound for $S_1 \cup S_2$ with $Q = 0$. Set $S = S_1 \sqcup S_2$. We now construct $T_S \succ 0$ with $Q = 0$. Each branch has vector span of dimension $k < m$, by Lemma 3.4. Apply the coefficient construction (23) to the bound already established on each branch. Since there is a vector perpendicular to all original vectors in that branch, Step 1 gives matrices satisfying (24). Write $X_i^*, W_i^*, T_i, C_i, Z_i$ for the data on S_i. Lemma 3.3 shows that the external forces on branch S_i span exactly U_{S_i}, a space of dimension k. Their row matrix is $P_i(W_i^*)^\top$. Since P_i has full column rank, W_i^* has rank k and its column space is U_{S_i}. The alignment $W_i^* = X_i^* T_i$ from (26), together with $T_i \succ 0$, gives

$$r_i = k, \qquad \operatorname{range} X_i^* = U_{S_i}. \tag{38}$$

By (20), the square matrix $X_0^* = [X_1^*\ X_2^*] \in \mathbb{R}^{2k\times 2k}$ is invertible. Thus the columns of X_1^* and X_2^* together form a basis of $\mathbb{R}^{2k}$. This is the fact that will allow us to prove $Q = 0$.

The cut between S and S_3 is the same cut viewed from opposite sides. For $A[S_3, S] = P_3 Q_3^\top$, choose the factors for S as $P_S = Q_3$ and $Q_S = P_3$, and set

$$T_S = T_3^{-1} \succ 0. \tag{39}$$

Then

$$X_S^* = W_3^* = X_3^* T_3, \qquad W_S^* = X_3^*.$$

Hence $W_S^* = X_S^* T_S$ for $T_S = T_3^{-1} \succ 0$. As in Steps 1–3, write $P_S = P_0 H$ and form J from T_1, T_2 and the interaction between S_1 and S_2. As in (37), first-order stationarity gives

$$J(X_0^*)^\top = H(W_S^*)^\top = HT_S H^\top (X_0^*)^\top.$$

The invertibility of X_0^* forces

$$J - HT_S H^\top = 0. \tag{40}$$

Thus $Q = 0$ for every movement. Define

$$C_S = \operatorname{diag}(C_1, C_2), \qquad Z_S = \operatorname{diag}(Z_1, Z_2) H (H^\top H)^{-1}.$$

The inverse exists because H has full column rank. The child identities, (40), and direct multiplication verify every condition in (24). The bound follows from (28) and (29), completing the induction for $k \geq 2$.

Finally, any other full-column-rank factorization of the same cut has the form $P_S B, Q_S B^{-\top}$ for an invertible matrix B. Under this change, X_S becomes $X_S B$. Replacing T_S by $B^{-1} T_S B^{-\top}$ preserves positive definiteness and leaves the quadratic bound unchanged. The auxiliary relations are unchanged after replacing Z_S by $Z_S B^{-\top}$ and leaving C_S fixed. Thus the claim holds for every full-column-rank cut factorization. □

4. Main proof of Theorem 1.1

Let V^* satisfy the assumptions of Theorem 1.1, and fix an arbitrary feasible configuration V. We prove that $E(V) - E(V^*) \geq 0$. If $n = 1$, the energy is constant, so assume $n \geq 2$. By Lemma 2.3, every nontrivial cut is nonzero.

Choose the rooted rank-decomposition used in Lemma 3.2. Let L and R be the sets of vertices corresponding to the leaves below the two children of the root. Then $[n] = L \sqcup R$. Moving both sets therefore changes the entire configuration from V^* to V. Lemma 3.1 gives

$$E(V) - E(V^*) = \Delta E_L + \Delta E_R - \sum_{i \in L,\, j \in R} A_{ij} \langle \Delta v_i, \Delta v_j \rangle.$$

Lemma 3.2 gives lower bounds for ΔE_L and ΔE_R. It remains to show that their sum is at least the cross term in this identity. We first do this for rank-width one, where the two coefficients are positive scalars, and then for $k \geq 2$.

Proof for $k = 1$. Step 1. We write the two bounds using the same cut factorization. Since $A[L, R]$ has rank one, write $A[L, R] = pq^\top$, and define

$$u = \sum_{i \in L} p_i v_i, \qquad w = \sum_{j \in R} q_j v_j.$$

Write u^*, w^* for these sums evaluated at V^*, and set $\Delta u = u - u^*$ and $\Delta w = w - w^*$. By Lemma 2.5, both original aggregates are nonzero and $w^* = \lambda u^*$ for some $\lambda > 0$. Viewed from R, the same relation is $u^* = \lambda^{-1} w^*$. The rank-one case of Lemma 3.2 therefore gives

$$\Delta E_L \geq \frac{\lambda}{2} \|\Delta u\|^2, \qquad \Delta E_R \geq \frac{1}{2\lambda} \|\Delta w\|^2.$$

Step 2. We substitute the bounds and complete the square. The factorization $A_{ij} = p_i q_j$ gives

$$\sum_{i \in L,\, j \in R} A_{ij} \langle \Delta v_i, \Delta v_j \rangle = \left\langle \sum_{i \in L} p_i \Delta v_i, \sum_{j \in R} q_j \Delta v_j \right\rangle = \langle \Delta u, \Delta w \rangle.$$

Substituting this expression and the two bounds into the total energy change yields

$$E(V) - E(V^*) \geq \frac{\lambda}{2}\|\Delta u\|^2 + \frac{1}{2\lambda}\|\Delta w\|^2 - \langle \Delta u, \Delta w\rangle$$
$$= \frac{1}{2}\|\sqrt{\lambda}\,\Delta u - \lambda^{-1/2}\Delta w\|^2 \geq 0.$$

Since V was arbitrary, V^* is globally optimal. □

Proof for $k \geq 2$. We distinguish whether the original vectors span $\mathbb{R}^m$. If they do not, a common perpendicular direction lets us test the second-order condition at the root. If they do, the coefficients constructed for the two root children are inverses, so we can complete a square as in the rank-one case.

Case 1: The original vectors do not span $\mathbb{R}^m$. Thus $\operatorname{rank} V^* < m$. Fix a unit vector e perpendicular to every v_i^*, and insert the root on any decomposition edge.

Step 1. We express the sum of the bounds minus the cross term as a quadratic form. Lemma 3.2 gives the bounds for the two root children L, R. Apply the coefficient construction (23), using e, to each of these bounds. For $I = L, R$, we obtain $T_I \succ 0$, $C_I \succeq 0$, and Z_I with

$$M[I, I] = C_I + P_I T_I P_I^\top, \qquad P_I^\top Z_I = I, \qquad C_I Z_I = 0,$$

and

$$\Delta E_I \geq \frac{1}{2}\operatorname{tr}(\Delta X_I T_I \Delta X_I^\top).$$

Here $M = \operatorname{diag}(\mu) - A$ is the matrix in Lemma 2.2. The matrices C_I, Z_I will allow us to test these coefficients using the second-order condition.

Choose Γ so that $A[L, R] = P_L \Gamma P_R^\top$, as in Lemma 2.4. Then the cross term is $\operatorname{tr}(\Delta X_L \Gamma \Delta X_R^\top)$. Consequently, the two bounds and the energy split give

$$E(V) - E(V^*) \geq \frac{1}{2}\operatorname{tr}(\Delta X_L T_L \Delta X_L^\top) + \frac{1}{2}\operatorname{tr}(\Delta X_R T_R \Delta X_R^\top) - \operatorname{tr}(\Delta X_L \Gamma \Delta X_R^\top)$$
$$= \frac{1}{2}\operatorname{tr}\Big([\Delta X_L\ \Delta X_R] J_{\text{root}} [\Delta X_L\ \Delta X_R]^\top\Big),$$

where

$$J_{\text{root}} = \begin{pmatrix} T_L & -\Gamma \\ -\Gamma^\top & T_R \end{pmatrix}.$$

Thus it suffices to prove $J_{\text{root}} \succeq 0$.

Step 2. We prove $J_{\text{root}} \succeq 0$ using the second-order condition. Group the child matrices into block diagonal matrices:

$$P_0 = \operatorname{diag}(P_L, P_R), \qquad C_0 = \operatorname{diag}(C_L, C_R), \qquad Z_0 = \operatorname{diag}(Z_L, Z_R).$$

Order the vertices with L first and R second. The identities in Step 1 and $M[L, R] = -A[L, R]$ give

$$M = C_0 + P_0 J_{\text{root}} P_0^\top, \qquad P_0^\top Z_0 = I, \qquad C_0 Z_0 = 0.$$

Let $x \in \mathbb{R}^{r_L + r_R}$ be arbitrary and put $z = Z_0 x$. These identities imply $P_0^\top z = x$ and $C_0 z = 0$, so

$$z^\top M z = x^\top J_{\text{root}} x.$$

To test the left-hand side, set $\xi_i = z_i e$ at every vertex. Since $e \perp v_i^*$, each ξ_i is tangent to the unit sphere at v_i^*. Applying the second-order condition to the matrix Ξ whose rows are $\xi_i^\top$, and using $\|e\| = 1$, gives

$$0 \leq \operatorname{tr}(\Xi^\top M \Xi) = z^\top M z = x^\top J_{\text{root}} x.$$

As x was arbitrary, $J_{\text{root}} \succeq 0$. Returning to the inequality in Step 1, we conclude that

$$E(V) - E(V^*) \geq \frac{1}{2}\operatorname{tr}\Big([\Delta X_L\ \Delta X_R] J_{\text{root}} [\Delta X_L\ \Delta X_R]^\top\Big) \geq 0. \tag{41}$$

The last inequality holds because the trace is the sum of the nonnegative quadratic forms obtained from the rows of $[\Delta X_L\ \Delta X_R]$.

Case 2: The original vectors span $\mathbb{R}^m$. Thus $\operatorname{rank} V^* = m$. Lemma 3.4 gives $m = 2k$ and the three sets S_1, S_2, S_3 in Figure 1. The root is placed so that its two children correspond to $S = S_1 \cup S_2$ and S_3.

Step 1. We identify the coefficients for the two root children. Step 4 of the proof of Lemma 3.2 establishes the bound on each of the three branches, including S_3. Step 5 constructs the bound for S, with coefficient $T_S = T_3^{-1}$ as in (39). Therefore,

$$\Delta E_S \geq \frac{1}{2}\operatorname{tr}(\Delta X_S T_3^{-1} \Delta X_S^\top), \qquad \Delta E_{S_3} \geq \frac{1}{2}\operatorname{tr}(\Delta X_3 T_3 \Delta X_3^\top).$$

To see how the cross term is expressed in these same aggregates, recall the factors chosen in Step 5: $A[S_3, S] = P_3 Q_3^\top$ and $A[S, S_3] = Q_3 P_3^\top$. Hence $\Delta X_S = (\Delta V_S)^\top Q_3$ and $\Delta X_3 = (\Delta V_{S_3})^\top P_3$, so

$$\sum_{i \in S,\, j \in S_3} A_{ij} \langle \Delta v_i, \Delta v_j \rangle = \operatorname{tr}(\Delta X_S \Delta X_3^\top).$$

Step 2. We substitute the bounds and complete the square. Since $[n] = S \sqcup S_3$, Lemma 3.1 gives

$$E(V) - E(V^*) = \Delta E_S + \Delta E_{S_3} - \operatorname{tr}(\Delta X_S \Delta X_3^\top).$$

Substitution of the two bounds yields

$$\begin{aligned} E(V) - E(V^*) &\geq \frac{1}{2}\operatorname{tr}(\Delta X_S T_3^{-1} \Delta X_S^\top) + \frac{1}{2}\operatorname{tr}(\Delta X_3 T_3 \Delta X_3^\top) - \operatorname{tr}(\Delta X_S \Delta X_3^\top) \\ &= \frac{1}{2}\|\Delta X_S T_3^{-1/2} - \Delta X_3 T_3^{1/2}\|_F^2 \geq 0. \end{aligned} \tag{42}$$

Here $T_3^{1/2}$ is the symmetric positive definite square root of T_3. Expanding the squared norm gives exactly the two quadratic terms and the cross term above.

Thus $E(V) \geq E(V^*)$ in both cases. Since V was an arbitrary feasible configuration, this proves Theorem 1.1. □

References

[BM03] S. Burer and R. D. C. Monteiro. A nonlinear programming algorithm for solving semidefinite programs via low-rank factorization. *Mathematical Programming*, 95:329–357, 2003. doi:10.1007/s10107-002-0352-8.

[BVB20] N. Boumal, V. Voroninski, and A. S. Bandeira. Deterministic guarantees for Burer–Monteiro factorizations of smooth semidefinite programs. *Communications on Pure and Applied Mathematics*, 73(3):581–608, 2020. doi:10.1002/cpa.21830.

[OS06] S.-I. Oum and P. Seymour. Approximating clique-width and branch-width. *Journal of Combinatorial Theory, Series B*, 96(4):514–528, 2006. doi:10.1016/j.jctb.2005.10.006.

[Oum12] S.-I. Oum. Rank-width and well-quasi-ordering of skew-symmetric or symmetric matrices. *Linear Algebra and its Applications*, 436(7):2008–2036, 2012. doi:10.1016/j.laa.2011.09.027.

[WW20] I. Waldspurger and A. Waters. Rank optimality for the Burer–Monteiro factorization. *SIAM Journal on Optimization*, 30(3):2577–2602, 2020. doi:10.1137/19M1255318.

[Wu] H. Wu. *Simple Cuts, Simple Landscapes: Global Synchronization and Rank-Width-One Matrices.* Manuscript.